

RITIKA SINGLA

Senior SDET / QA Automation Architect • Playwright (TypeScript) • AI/LLM Test Engineering • API & CMS/AEM Automation • CI/CD

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SUMMARY

QA Automation Architect and hands-on SDET with 14+ years building test automation for distributed, cloud-native microservices at Maersk, IBM, and IHS Markit (S&P Global). Hands-on with Playwright (TypeScript) and Selenium 4/Java for scalable, framework-first UI automation, and deep REST API and contract-testing expertise (RestAssured, Postman). Practitioner of AI-assisted test engineering — building deterministic, repeatable validation harnesses with human-in-the-loop review, and using AI coding assistants (Claude Code, GitHub Copilot) for test generation, coverage-gap analysis, and failure triage. Experience testing AEM/CMS components, content workflows, and publishing pipelines. Leads a 5-member QA/SDET team, setting technical direction and mentoring engineers. Strong CI/CD and observability background (GitHub Actions, Jenkins, Grafana/Loki) across regulated financial-services environments. ISTQB and AWS Cloud Practitioner certified.

CORE SKILLS

- **UI automation:** Playwright (TypeScript) — hands-on: framework design, Page Object Model, parallel execution, auto-wait/tracing; Selenium 4 / Java (expert)
- **API & contract testing:** RestAssured, Postman, Pytest, Allure, Serenity — status/schema/payload validation, chained request flows, contract tests
- **AI / LLM test engineering:** AI-assisted test generation (Claude Code, GitHub Copilot); deterministic, repeatable validation harnesses; human-in-the-loop review; coverage-gap analysis and automated failure triage
- **CMS / AEM:** Adobe Experience Manager component & workflow testing, content validation, and publishing-pipeline verification across CMS-driven applications
- **Mocking & service virtualization:** WireMock, Microcks — stub authoring and mock infrastructure enabling parallel team development
- **CI/CD & DevOps:** GitHub Actions, Jenkins — automated quality gates, tiered regression suites, environment consistency; Docker
- **Performance & reliability:** JMeter — latency (p50/p95), throughput, and scalability testing across APIs and services
- **Microservices & cloud:** AWS (ECS, Fargate, Lambda, Kinesis), Kafka, Temporal/Camunda — service-boundary and async-workflow testing
- **Data quality:** SQL/NoSQL validation, ETL testing, Kafka streaming pipeline verification
- **Ways of working:** Git/GitHub feature-branch workflow and PR reviews; Jira-traceable coverage; embedded Agile squads; shift-left

EXPERIENCE

Quality Assurance Architect (hands-on SDET) — A.P. Moller - Maersk | Dec 2023 - Present

- Lead a team of 5 QA engineers/SDETs — setting technical direction, running code reviews, planning coverage and delivery, and building team capability across the automation program.
- Own end-to-end automated testing for cloud-native microservices on AWS (ECS, Fargate, Lambda, Kinesis) — UI, REST APIs, async workflows, and Kafka streaming data.
- Architected API and contract test suites (RestAssured, Pytest) and validated WebSocket data and network latency across distributed services.
- Built Playwright (TypeScript) automation for next-generation UI frameworks alongside established Selenium 4 / Java suites; standardized framework patterns and parallel execution.
- Tested AEM/CMS components and content workflows — validating content authoring, workflow transitions, and publishing pipelines for CMS-driven applications.
- Built a deterministic, Python-based validation harness for production datafeeds using an AI coding-assistant (Copilot) agent architecture — AI confined to authoring and summarizing, with all pass/fail checks deterministic and human-reviewed.
- Introduced WireMock and Microcks service virtualization — authored stubs and mock infrastructure that unblocked parallel development for 4+ teams.
- Embedded automated quality gates in GitHub Actions pipelines, cutting production escapes by ~40%; used Grafana/Loki observability for fast root-cause analysis.
- Cut regression cycle time from ~5 days to under 8 hours with 80%+ coverage of critical flows; performed ETL and data-integrity validation across SQL, NoSQL, and Kafka pipelines.

Quality Assurance Lead (Senior Advisory Technical Specialist) — IBM | Nov 2021 - Dec 2023

- Designed Selenium and API automation frameworks standardizing test practice across enterprise client programs in regulated environments; expanded regression coverage.

- Integrated automated testing into client CI/CD workflows (Jenkins) with development and DevOps teams, reducing release-cycle bottlenecks.
- Defined test strategy, effort estimation, and automation-adoption roadmaps; mentored 10+ engineers on automation best practice, test design, and tooling.

Senior Quality Assurance Engineer — IHS Markit (now S&P Global) | May 2014 – Nov 2021

- Built and maintained Selenium/Java automation frameworks across multiple financial-data product lines in a regulated domain, expanding automated regression coverage year over year.
- Executed full-stack testing across UI, APIs, SQL/NoSQL databases, and ETL pipelines, safeguarding data integrity for financial-services clients.
- Managed test code in GitHub/SVN; contributed to CI/CD pipeline configuration for automated test execution.

Test Engineer — Infosys | Oct 2011 – Feb 2014

- Manual and automation testing across enterprise client projects.

SELECTED ACHIEVEMENTS

- Built 7+ reusable automation libraries/utilities (auth handling, test-data factories, API clients, custom reporters) reused across multiple product teams.
- Reduced production defect leakage by 60% through shift-left practices — API contract checks and component-level tests running pre-merge.
- Established quality KPIs and dashboards (coverage, pass rate, escape rate, MTTR) giving leadership clear release-readiness visibility.

CERTIFICATIONS & EDUCATION

- ISTQB Certified Tester · AWS Certified Cloud Practitioner
- B.Tech, Computer Engineering — Chandigarh Engineering College (2007–2011)